Project Documentation
Strategic Product Placement Analysis: Unveiling Sales Impact with Tableau Visualization
(Data Analytics with Tableau)

Team Details

- **Team ID**: **LTVIP2026TMIDS36203**
- **Team Size**: 3
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1. INTRODUCTION

Project Overview

Retail businesses generate large volumes of sales data across multiple product categories, regions, and time periods. However, this data is often stored in spreadsheets and analyzed manually, making it difficult to identify patterns, measure product placement impact, and make timely strategic decisions. Manual reporting methods lack interactivity, real-time visualization, and structured insights.

This project aims to develop an interactive Tableau dashboard that analyzes product placement and its impact on sales performance. The system transforms raw sales data into meaningful visual insights using KPIs, trend analysis, and product-level comparisons. The dashboard provides a centralized and user-friendly interface for better business decision-making.

Purpose

The purpose of this project is to replace static Excel-based analysis with a dynamic and visual analytics solution using Tableau. The dashboard enables real-time performance tracking, accurate KPI calculation, structured data visualization, and improved product placement strategy. It ensures better clarity, faster insights, and scalable business intelligence.

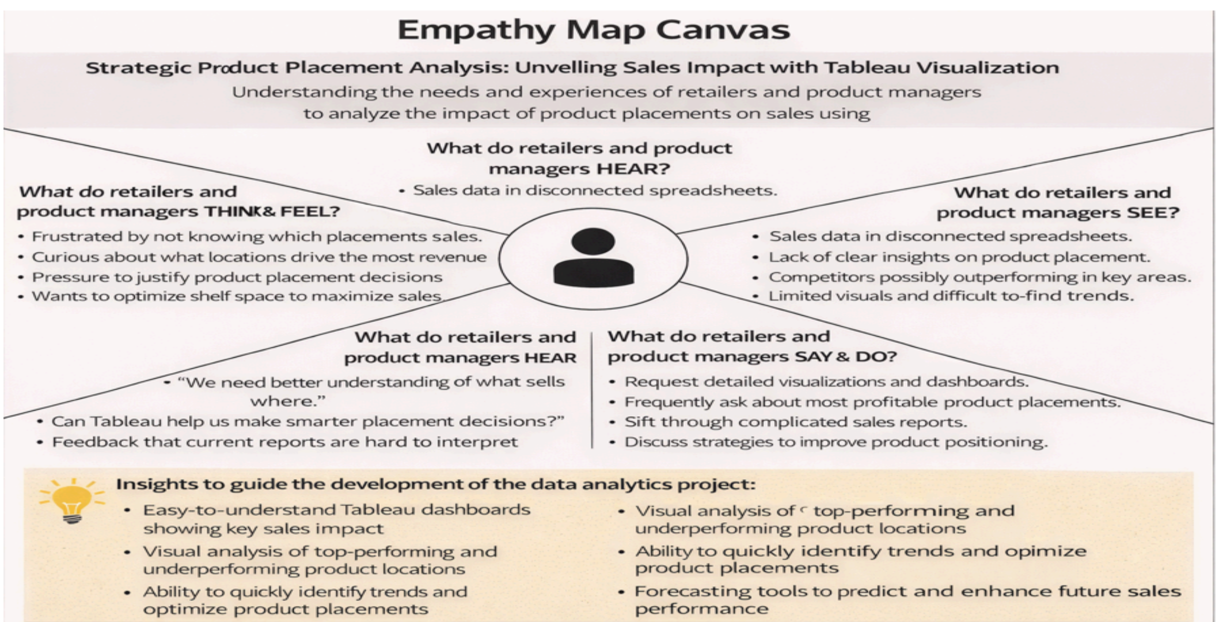
2. IDEATION PHASE

Problem Statement

Retail managers face difficulty understanding how product placement affects sales due to fragmented datasets and non-interactive reports. Manual analysis leads to delayed insights, errors in calculations, and lack of visibility into product performance. Without proper visualization tools, identifying top-performing and underperforming products becomes challenging.This project addresses these issues by creating a centralized Tableau dashboard that provides structured, interactive, and real-time sales analysis.

Empathy Map Canvas

● **Who?** Retail managers, sales analysts, business owners
● **What do they think and feel?** Need quick, accurate insights for better decisions
● **What do they see?** Multiple Excel sheets and inconsistent reports
● **What do they say and do?** Frequently request updated performance reports
● **What do they hear?** Concerns about declining sales or poor product placement
● **Pain?** Slow analysis, confusion, missed revenue opportunities
● **Gain?** Clear visualization, better planning, improved profitability



Brainstorming

The team explored multiple approaches:

● Continuing manual Excel reporting
● Using static reporting dashboards
● Implementing an interactive Tableau dashboard

The Tableau solution was selected due to its powerful visualization features, dynamic analytics, user-friendly interface, and scalability.

3. REQUIREMENT ANALYSIS

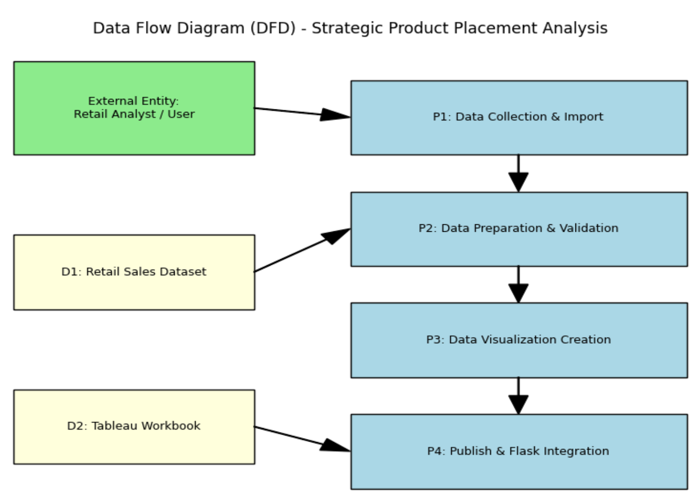
Customer Journey Map

The user opens the Tableau dashboard to review business performance.
The dashboard immediately displays KPIs such as Total Sales and Profit %.
The user analyzes category-wise sales and observes time-based trends.
Product-level performance charts help identify top-performing items.
Insights gained from the dashboard support strategic product placement decisions.

Solution Requirements

● Centralized interactive dashboard
● Display of Total Sales, Profit %, Growth %
● Category-wise sales visualization
● Product-level performance analysis
● Time-based sales trend chart
● Accurate data source connection and refresh

Data Flow Diagram



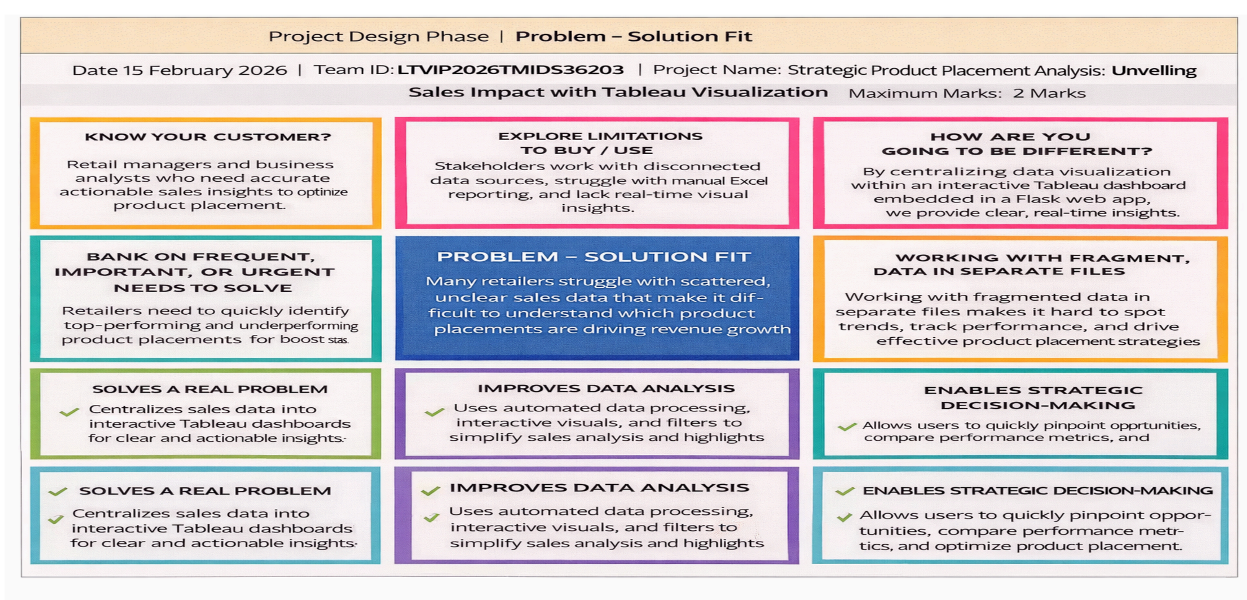
Technology Stack

● Tool: Tableau Desktop
● Data Source: Excel / CSV dataset
● Features Used: Calculated Fields, KPIs, Trend Analysis
● Visualization Types: Bar Chart, Line Chart, KPI Cards

4. PROJECT DESIGN

Problem–Solution Fit

The solution directly addresses the inefficiencies of manual reporting by introducing an interactive visualization platform. The Tableau dashboard ensures structured data representation, accurate performance measurement, and better product placement evaluation. By converting raw data into clear visual insights, the system improves business decision-making and operational efficiency.



Proposed Solution

A comprehensive Tableau dashboard was developed featuring:

- KPI Cards displaying Total Sales and Profit %
- Category-wise Sales Bar Chart
- Product-Level Performance Analysis
- Time-Based Sales Trend Line Graph
- Structured and validated data source integration

Solution Architecture

The architecture consists of four layers:

1. Presentation Layer

Interactive Tableau dashboard displaying KPIs and visual analytics.

2. Logic Layer

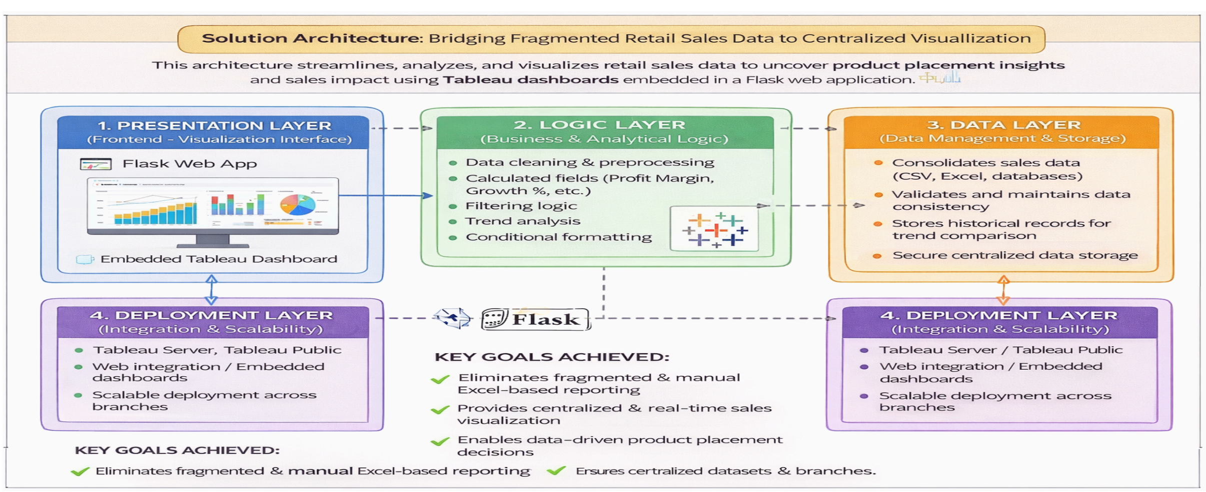
Calculated fields, aggregation functions, and trend analysis logic.

3. Data Layer

Structured Excel/CSV dataset connected to Tableau.

4. Deployment Layer

Dashboard sharing through Tableau platform for stakeholder access.

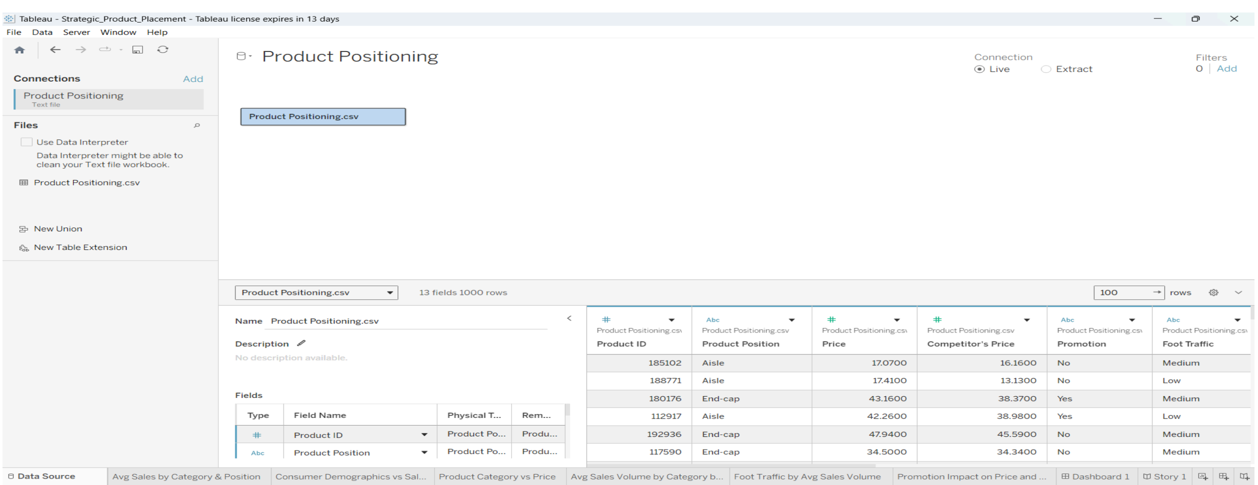


5. PROJECT PLANNING & SCHEDULING

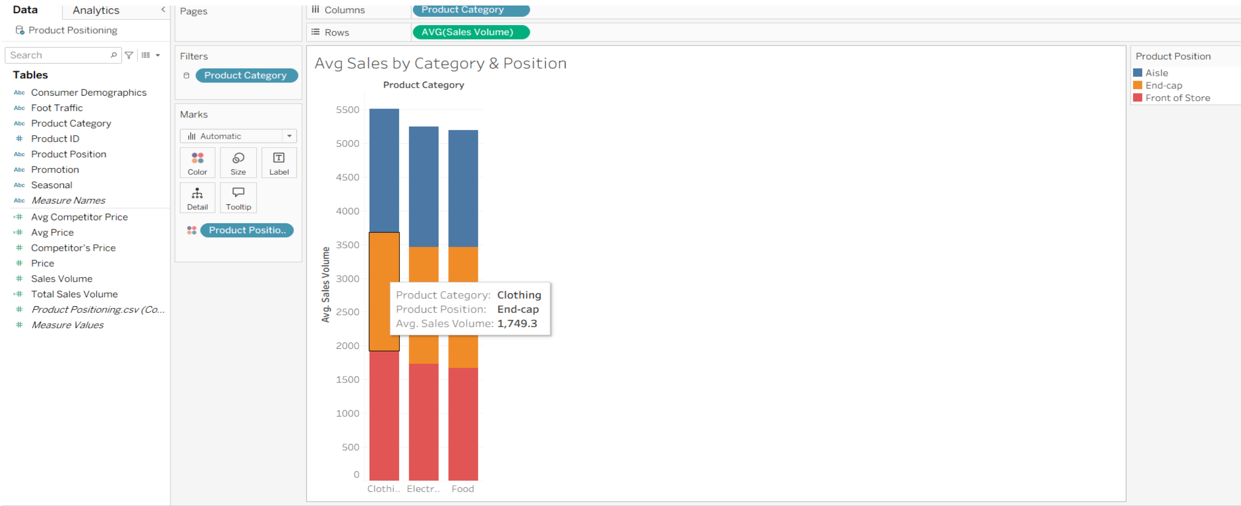
1. Data Collection
2. Data Cleaning & Preparation
3. Data Import into Tableau
4. KPI Creation using Calculated Fields
5. Visualization Development
6. Dashboard Design & Layout
7. Testing & Validation

6. IMPLEMENTATION WORKFLOW

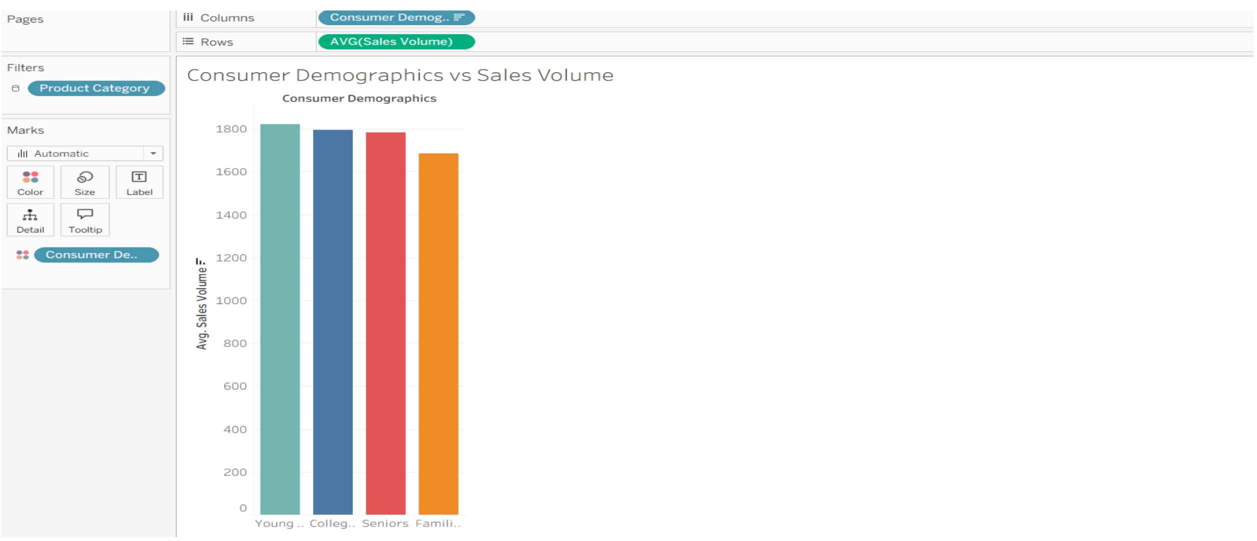
- Import dataset into Tableau



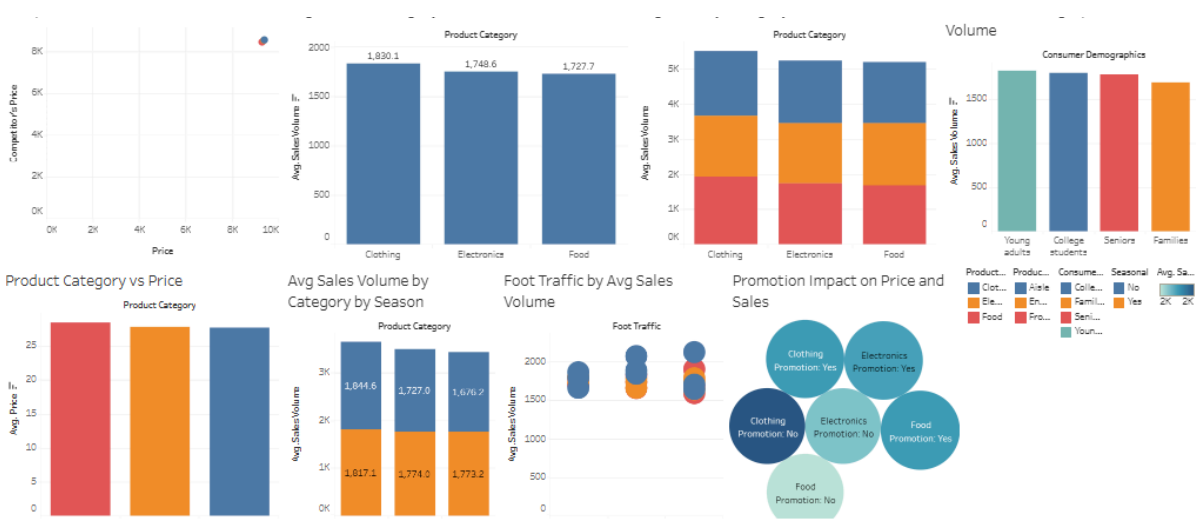
- Verify data types and field structure



- Create calculated fields (Profit %, Growth %)
- Develop KPI cards
- Create bar chart for category-wise sales



- Create line chart for trend analysis
- Combine worksheets into final dashboard



- Validate accuracy of all metrics

7. FUNCTIONAL AND PERFORMANCE TESTING

Functional Testing

Functional testing was conducted to verify that the Flask web application and Tableau dashboard integration worked correctly.

1. Flask Application Execution Testing

The Flask framework was successfully installed and configured.
The application was executed using the command:

python app.py

After running the command, the Flask development server started successfully and generated a local URL (http://127.0.0.1:5000/). When accessed through the browser, the web application loaded without errors.

This confirms that the backend server configuration is functioning properly.

2. HTML Template Integration Testing

The project structure includes:

- `app.py` (Flask backend file)
- `templates/index.html` (Frontend HTML file)

The `index.html` file was properly linked using Flask's `render_template()` function. Upon running the server, the HTML page loaded correctly in the browser, displaying the project title and layout.

This confirms that the frontend and backend integration is successful.

3. Tableau Public Embed Integration Testing

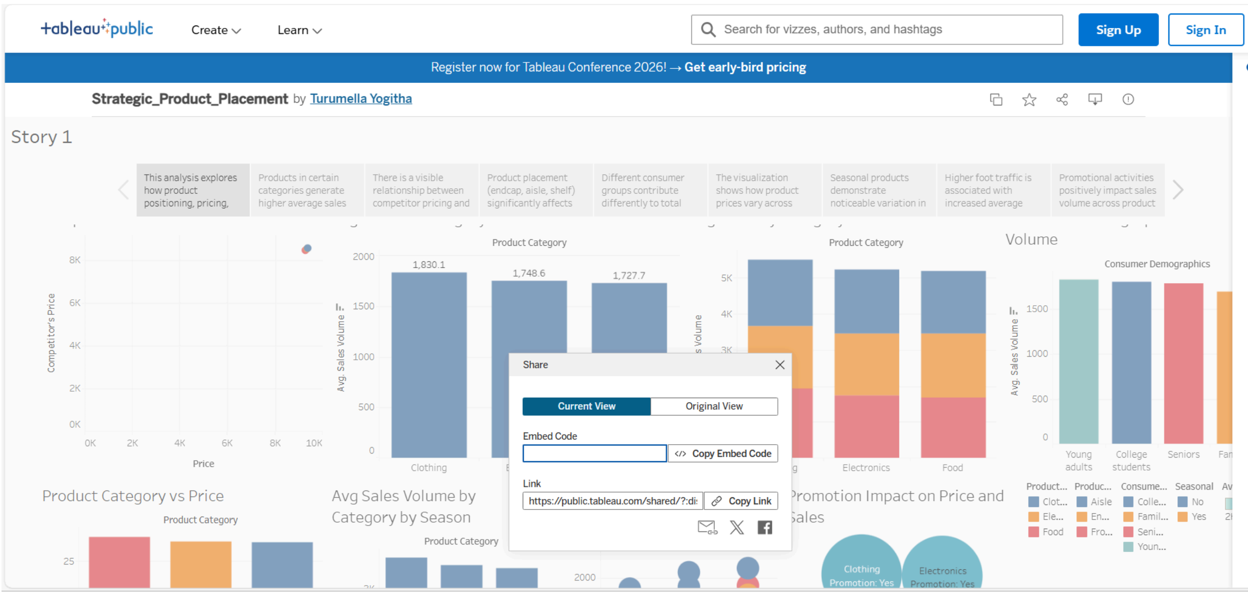
The Tableau dashboard was first published to Tableau Public.
After publishing:

1. The **Share** option was selected.
2. The **Embed Code** was copied.
3. The embed code was pasted inside the `<div>` section of `index.html`.

After restarting the Flask server and refreshing the browser:

- The Tableau dashboard loaded successfully inside the web application.
- All visualizations were visible.
- No broken links or script errors occurred.

This confirms successful integration between Tableau Public and the Flask web application.



Performance Testing

Performance testing was conducted to evaluate system responsiveness and stability.

1. Server Response Testing

- Flask server starts within a few seconds.
- The application loads quickly in the browser.
- No server crash or timeout occurred during execution.

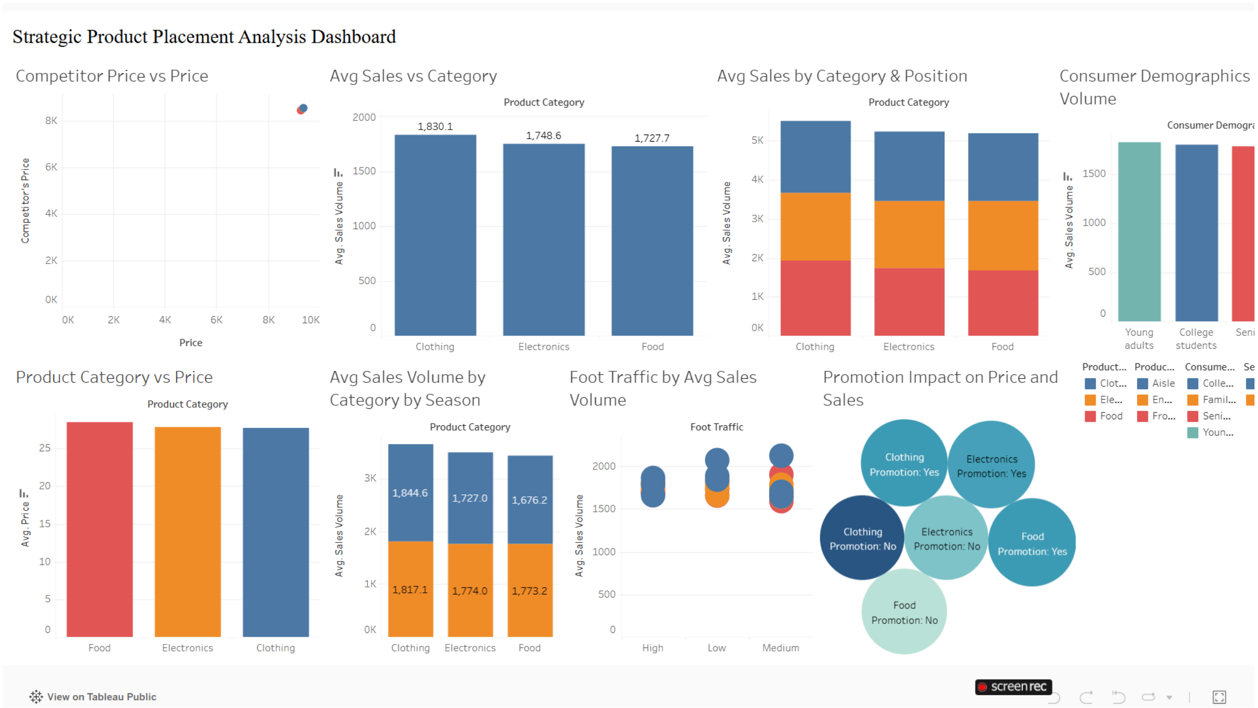
2. Dashboard Rendering Testing

- The embedded Tableau dashboard loads completely.
- All charts and KPIs render correctly.
- No distortion or alignment issues were observed.

3. Stability Testing

The application was refreshed multiple times to test consistency.
Results observed:

- The dashboard reloads without errors.
- No delay in visualization rendering.
- Stable connection with Tableau Public.



8. ADVANTAGES & DISADVANTAGES

Advantages

- Real-time interactive analysis
- Improved strategic decision-making
- Clear visualization of product performance
- Eliminates manual calculation errors
- Scalable for future business growth

Disadvantages

- Requires Tableau knowledge for maintenance
- Dependent on clean and accurate dataset

9. CONCLUSION

The project successfully implements an interactive Tableau dashboard to analyze product placement and sales impact. By transforming raw data into structured visual insights, the system enhances business intelligence, improves decision-making efficiency, and supports strategic planning. The dashboard eliminates the limitations of manual reporting and ensures accurate performance tracking.

10. FUTURE SCOPE

- Integration with inventory management data
- Predictive sales forecasting
- Multi-branch comparative dashboards
- Advanced analytical metrics and automation